Free-fall lab

antocciphysics | TEACHER KEY | CPA

Learning target: Estimate g from measured motion and discuss measurement limits.

Suggested first pass: 1, 2, 3, 7. Add independent, mixed and stretch work from the bank as time and student evidence permit. Estimated times are planning estimates, not completion requirements.

01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 5; 3 course days later.*
3 N, attractive toward the other mass.

02. Explain a worked example (worked)
Points 8, 10, 12 with mean 10; timing scale or calibration error could shift all readings.

03. Look and explain (model)
(5-1)/(.5-.1)=10 m/s² downward.

04. Build the solution (practice)
Mean 10 m/s²; range 9-11. Variation helps estimate uncertainty and review technique.

05. Use the idea (reason)
Controlled release/setup; height/time orv data; pushed release changes initial condition. Preserve flagged trial and explain exclusion if justified.

06. Choose a method (mixed) *Review: Unit 1, Day 18; 12 course days later.*
14 N left; 18-4-14=0 N.

07. Exit check (exit)
10 m/s²; timing/sensor fit/release etc.

08. STRETCH: Apply it another way (stretch)
Both means 10; first group smaller spread and better repeatability. Accuracy needs a reference and systematic-error review.

Teaching note: Investigation companion. Collect actual trials only with teacher-approved equipment; otherwise label all work supplied-data analysis.

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01. Bring back an earlier idea (retrieve) *Review: Unit 2, Day 5; 3 course days later.*
$F \approx 2.401 \times 10^{-5}$ N, attractive. The incorrect numerical answer is 5 times too large when SI values are used.

02. Explain a worked example (worked)
Points 8, 10, 12 with mean 10; timing scale or calibration error could shift all readings.

03. Represent and interpret (model)
(5-1)/(.5-.1)=10 m/s² downward.

04. Solve and justify (practice)
Mean 9.9; half-range .3 m/s²; deviation≈1.02%. Half-range is a simple spread estimate, not automatically a confidence interval.

05. Analyze the reasoning (reason)
Controlled release/setup; height/time orv data; pushed release changes initial condition. Preserve flagged trial and explain exclusion if justified.

06. Choose a method (mixed) *Review: Unit 1, Day 18; 12 course days later.*
At rest T=39.2 N. Accelerating: T=m(g+a)=51.2 N.

07. Exit check (exit)
9.8 m/s²; multiple points reduce sensitivity to individual noise and reveal trend.

08. STRETCH: Apply it another way (stretch)
Mean 9.8, half-range 0.4 m/s², deviation≈0.102%. No; a shared calibration bias can give consistent but inaccurate values.

Teaching note: Investigation companion. Collect actual trials only with teacher-approved equipment; otherwise label all work supplied-data analysis.